

AeroGrid Engineering Solutions Package

Artefact 1: Data Processing Code

```
import pandas as pd

data = pd.read_csv("telemetry_data.csv")

summary = data.groupby("turbine_id").agg({
    "temperature": "mean",
    "vibration": "max"
}).reset_index()

failing = summary[(summary["temperature"] > 85.0) | (summary["vibration"] > 15.0)]

print("Failing Turbines:")
print(failing["turbine_id"].tolist())
```

Artefact 2: Dockerfile

```
FROM python:3.9-slim
WORKDIR /app
COPY telemetry_data.csv .
COPY script.py .
RUN pip install pandas
CMD ["python", "script.py"]
```

Artefact 3: System Architecture

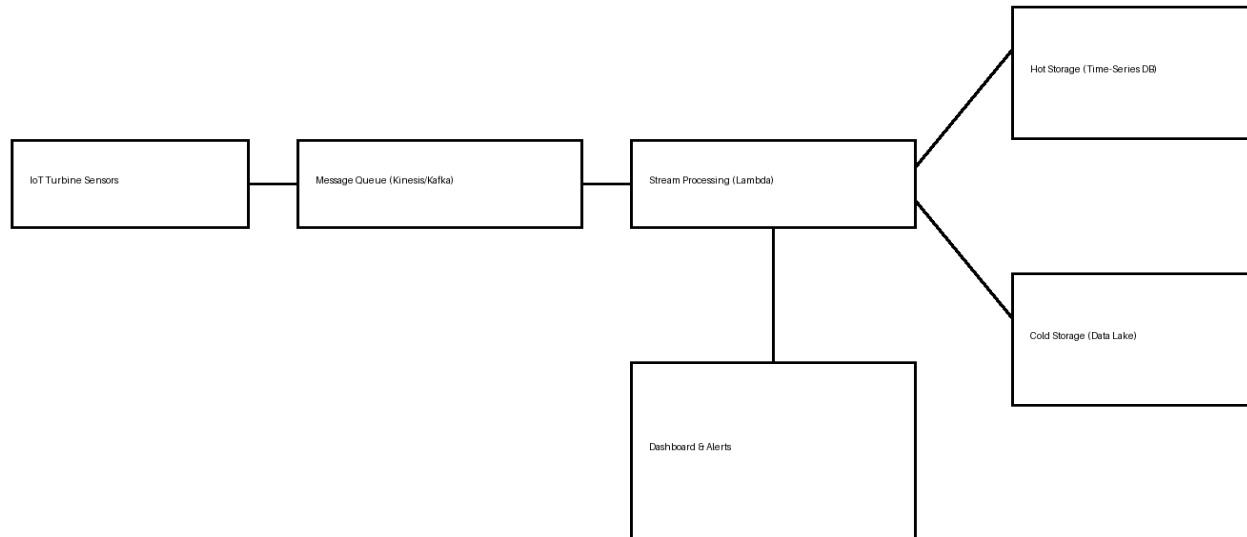


Figure 1: Scalable Cloud Architecture for Real-Time Turbine Monitoring

Artefact 4: Engineering Report

This system analyses turbine telemetry data to identify early signs of failure using clearly defined anomaly thresholds. Turbines with average temperatures exceeding 85°C or vibration levels above 15 mm/s are flagged as at-risk and require urgent maintenance intervention. This ensures that faults are detected before they escalate into costly breakdowns. To address the limitations of the current legacy infrastructure, a modern cloud-based architecture has been proposed. Data generated from IoT sensors is first transmitted into a message queue such as Apache Kafka or AWS Kinesis. This layer acts as a buffer, preventing data loss during spikes and ensuring reliable ingestion. The data is then processed in real time using stream processing tools such as AWS Lambda, allowing anomaly detection to occur instantly as new data arrives. The system separates storage into hot and cold layers. Hot storage uses a fast time-series database for real-time dashboards and monitoring, while cold storage uses a data lake for long-term analytics at a lower cost. This improves scalability, reliability, and performance while reducing operational risks. Additionally, cost optimisation is achieved through lifecycle policies that automatically move older data into cheaper storage tiers, ensuring efficient use of cloud resources.